Sample Solutions: IT Maths Jan exam 2021

Question 1 (Attempt All parts) 40 Marks.

a) $A = \begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ $C = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$

i) Size of $A = 3 \times 3$
rows x columns.

Size of $B = 3 \times 1$

ii) $a_{23} = 1$ $b_{11} = 1$
row, col

iii) $A^T(2B+C)$ — Transpose (rows become cols, cols become rows)

$A^T = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ $2B = 2\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$

$A^T(2B+C) =$

$$\begin{bmatrix} 1 & 3 & 1 \\ 3 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}\left(\begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}\left(\begin{bmatrix} 3 \\ 5 \\ 5 \end{bmatrix}\right)$$

3x3 · 3x1

$$= \begin{bmatrix} 1\times3+3\times5+1\times5 \\ 3\times3+0\times5+1\times5 \\ 2\times3+1\times5+1\times5 \end{bmatrix} = \begin{bmatrix} 3+15+5 \\ 9+0+5 \\ 6+5+5 \end{bmatrix} = \begin{bmatrix} 23 \\ 14 \\ 16 \end{bmatrix}$$

b) Recursion

$t(1) = 2$

$t(n) = t(n-1) + 3 \quad (n > 1)$

$t(2) = t(2-1) + 3$

$= t(1) + 3$

$= 2 + 3$

$t(2) = 5$

$t(3) = t(3-1) + 3$

$= t(2) + 3$

$= 5 + 3$

$t(3) = 8$

$t(4) = t(4-1) + 3$

$= t(3) + 3$

$= 8 + 3$

$t(4) = 11$

c) Give a definition of Sampling

2 types of Sampling + how they work: Random Sampling
Stratified Random Sampling
(or others)

d) i) mean: $\bar{x} = \frac{\sum_i x_i}{n} = \frac{4.7 + 3.7 + 4.0 + 5.7 + 2.7 + 1.7 + 2.3}{7}$

$= \frac{24.8}{7}$

$\bar{x} = 3.54$ (rounded to 2 decimal places)

ii) standard deviation: $S^2 = \frac{\sum x_i^2 - n(\bar{x})^2}{n-1}$

$$S^2 = \frac{(4.7)^2+(3.7)^2+(4.0)^2+(5.7)^2+(2.7)^2+(1.7)^2+(2.3)^2 - 7(3.54)^2}{7-1}$$

$$= \frac{100.55 - 87.7212}{6}$$

$$S^2 = 2.138$$

$$S = \sqrt{2.138} = 1.46 \text{ (2 decimal places)}$$

e) i) $P[\text{SUM} > 5] = 26/36 = 13/18$

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

2 dice so total number of possible outcomes = 6×6=36

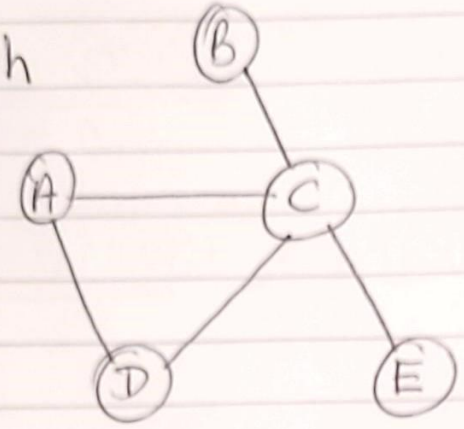
event space for SUM

ii) Permutations: $^nP_r = \frac{n!}{(n-r)!}$

$n = 7$, $r = 3$ $\rightsquigarrow$ $^7P_3 = \frac{7!}{(7-3)!} = \frac{7\times6\times5\times4\times3\times2\times1}{4\times3\times2\times1}$

$$^7P_3 = 210$$

f) i) Graph



ii) $V = \{A, B, C, D, E\}$ V = Vertices

$E = \{(A,C), (A,D), (B,C), (C,D), (C,E)\}$ E = Edges.

iii) Adjacency Matrix.

	A	B	C	D	E
A	0	0	1	1	0
B	0	0	1	0	0
C	1	1	0	1	1
D	1	0	1	0	0
E	0	0	1	0	0

g) gradient (slope) at $(3, 1)$ $\quad y = 2x^3 - 6x + 3$

get $\frac{dy}{dx} = (2)(3)x^{3-1} - (6)(1)x^{1-1} + 0$ constant goes to zero.

$= 6x^2 - 6$

sub in $x = 3$ $\rightarrow$ $6(3)^2 - 6$

$= 48$

gradient (slope) $= 48$

h) i) $\int (x^{-3}) . dx$ use power rule of integration.

$= \frac{x^{-3+1}}{(-3+1)} + C$ ← add 1 to the power; ← divide by new power; ← constant of integration.

$= \frac{x^{-2}}{-2} + C$

ii) $\int (6x^2 + 4x + 2) . dx$ use power rule of integration - apply to each term.

$= \frac{6x^{2+1}}{3} + \frac{4x^{1+1}}{2} + 2x + C$

divide by new power → 3; ← divide by new power; ↑ constant gets an x!

$= 2x^3 + 2x^2 + 2x + C$